

ALGORITHM THEORETICAL BASIS DOCUMENT

ATBD — Version 3.0

Annual Wetland Land Cover Classification from Landsat Time Series

Self-Supervised Approach | Dry-Season Median Compositing | RF Calibration | R Post-Processing Pipeline

Ibaji Floodplain, Nigeria — 1985–2024

Version	Date	Changes
1.0	2024	Initial release. BAP compositing. Multi-year loop. Uncertainty index (H+M) exported as raster.
2.0	2025	Median-of-indices compositing replaces BAP. Single-year processing mode. Fixed JM_YEAR parameter. arrayFlatten fix for MULTIPROBABILITY export. Uncertainty raster removed. Confidence map added. Per-class probabilities in validation CSV. Isotonic calibration in R.
3.0	2025	Added full R pipeline documentation (Modules 1–4). Expanded Section 9 with calibrated uncertainty (Shannon entropy formula) and explicit distinction from calibrated confidence. Stream B NoData bug fix documented (v2.2). pix_km2 geographic CRS fix documented (v2.3). AI assistance documentation added (Section 14).

Platform: Google Earth Engine (JS API) + R post-processing

Abstract

This document constitutes the Algorithm Theoretical Basis Document (ATBD) for an annual wetland land cover classification workflow applied to the Ibaji floodplain, Nigeria, covering the Landsat archive period 1985–2024. Version 3.0 extends v2.0 with a comprehensive description of the four-module R post-processing pipeline and an expanded treatment of probability calibration outputs.

The R pipeline consists of: Module 1 (isotonic regression calibration of RF probabilities, producing `calibrated_confidence` and `calibrated_uncertainty` per year); Module 2 (composite Quality Index combining coverage, confidence, balanced accuracy, and SAR availability); Module 3 (land cover change analysis via confidence-weighted temporal smoothing and uncertainty-gated change detection, with documented fixes for NoData encoding and geographic CRS unit handling); and Module 4 (variable importance aggregation across the 40-year series with pre/post-SAR comparison).

A dedicated section clarifies the distinction between calibrated confidence and calibrated uncertainty — two outputs derived from the same calibrated probability distribution but measuring fundamentally different aspects: the certainty of the assigned class vs. the dispersion of probability mass across all classes. The document concludes with a section documenting the use of AI assistance in workflow development, in compliance with emerging standards for AI disclosure in scientific publications.

1. Introduction and Scope

Tropical inland wetlands provide critical ecosystem services including flood buffering, carbon sequestration, and habitat for specialised biodiversity. Long-term systematic mapping of wetland dynamics in West Africa remains limited by the scarcity of field reference data and the complexity of multi-mission satellite archives.

This algorithm addresses both limitations through a self-supervised strategy: training labels are derived from ESA WorldCover 2020 automatically, and classification is applied independently to each year of the Landsat archive. The algorithm produces per-year outputs consisting of a land cover map, a confidence map, and a per-class probability image, with a four-module post-processing pipeline in R that calibrates RF class probabilities, assesses annual quality, analyses multi-decadal change, and aggregates variable importance.

The workflow is implemented as a publicly accessible GEE JavaScript script and a set of R modules controlled by a single configuration file. All processing is cloud-based (GEE) or local (R), with no proprietary software dependencies.

How to use this document

Sections 1–12 document the GEE portion of the algorithm. Sections 13–16 document the R pipeline. Section 17 documents known limitations. Section 18 contains the AI assistance declaration.

2. Data Inputs

2.1 Landsat Imagery

Mission	Sensor	Period	GEE Collection
Landsat 4	TM	1982–1993	LANDSAT/LT04/C02/T1_L2
Landsat 5	TM	1984–2013	LANDSAT/LT05/C02/T1_L2
Landsat 7	ETM+	1999–present	LANDSAT/LE07/C02/T1_L2
Landsat 8	OLI/TIRS	2013–present	LANDSAT/LC08/C02/T1_L2
Landsat 9	OLI-2/TIRS-2	2021–present	LANDSAT/LC09/C02/T1_L2

All Collection 2 Tier-1 Level-2 Surface Reflectance products. Tier-1 selection ensures only scenes that meet USGS geometric and radiometric standards are used.

2.2 Sentinel-1 SAR (post-2014)

Sentinel-1 IW GRD images (VV + VH polarisations) from the COPERNICUS/S1_GRD collection are integrated for years ≥ 2015 . The SAR backscatter distinguishes open water (specular reflection, low VV) from flooded vegetation (double-bounce, high VV/VH) — a discrimination that is physically inaccessible to optical sensors during the dry season. Three features are added: S1_VV, S1_VH, and S1_VVH_ratio, all as seasonal medians in dB.

⚠ Client-side S1 availability check

S1 availability is checked with a client-side year comparison (`year >= S1_MIN_YEAR`) rather than a server-side `ee.Algorithms.If()`. This is intentional: `ee.Algorithms.If()` returns an `ee.Image`

object regardless of the condition result, which would cause type errors when no S1 data is available. The client-side check returns a JavaScript false value, and the feature stack construction checks for this explicitly.

2.3 Ancillary Datasets

Dataset	Resolution	Variables
USGS SRTMGL1_003	30 m	Elevation, slope, roughness (90 m σ), TWI
MERIT Hydro v1.0.1	~90 m	Upstream drainage area (for TWI computation)
TerraClimate (IDAHO_EPSCOR)	~4.6 km	Annual mean: tmmn, tmmx, def, vap, soil

The Topographic Wetness Index ($TWI = \ln(A / \tan(\beta))$) is the most consistently important feature across all 32 processed years, ranking in the top-3 in every year regardless of Sentinel-1 availability. In a sub-humid floodplain with low topographic relief, TWI captures the position of each pixel in the drainage network, which is a strong proxy for inundation probability even when the surface is dry during the composite season.

3. User Configuration

The GEE script exposes a user configuration block at the top of the file. Only this block requires editing between runs. Table 3 documents all configurable parameters.

Variable	Default	Description
YEAR	2020	Year to process. Range: 1985–2024 (Landsat archive).
JM_YEAR	2020	Reference year for JM separability. MUST remain fixed across all years for a consistent time series.
AOI_ASSET	(user path)	GEE asset path of the study area polygon.
OUTPUT_FOLDER	(user name)	Google Drive output folder.
DRY_SEASON	{11,1,5,31}	Start (Nov 1) and end (May 31) of dry season window.
JM_BANDS	10 indices	Spectral indices used for JM separability analysis.
JM_THRESHOLD	1.8	JM distance threshold; pairs below are merge candidates.
TARGET_SAMPLES_PER_CLASS	150	Maximum balanced training sample size per class.
TRAIN_FRACTION	0.70	Training fraction of balanced sample (30% retained for validation).
Z_THRESHOLD	3	Spectral outlier removal: $ z > \text{threshold}$ excluded.
RF_PARAMS	(object)	Random Forest hyperparameters: 300 trees, 6 vars/split, minLeaf=2, bagFrac=0.7, maxNodes=128.
S1_MIN_YEAR	2015	First year with Sentinel-1 data. Client-side check.

4. Pre-Processing

4.1 Cloud and Quality Masking

For each Landsat scene, invalid pixels are identified and masked via two quality bands: QA_PIXEL (fill, dilated cloud, cloud shadow, snow, cirrus for L8/9) and QA_RADSAT (saturated band values).

4.2 Band Harmonisation

All missions are renamed to a common scheme: BLUE, GREEN, RED, NIR, SWIR1, SWIR2, THERMAL (optional). L4/5/7 use B1–B7; L8/9 use SR_B2–B7 + ST_B10.

4.3 Spectral Index Computation

i Critical design choice: indices computed before compositing

Indices are computed on each individual scene BEFORE aggregation. This ensures that each scene's bands are internally consistent (same atmosphere, same geometry) when the index is computed. The median is then taken across the per-scene index values, not across bands. Computing the median of bands first and then deriving indices from band medians produces different results, particularly for ratio indices.

Index	Category	Formula
MNDWI	Water	$(\text{GREEN} - \text{SWIR1}) / (\text{GREEN} + \text{SWIR1})$
AWEI	Water	$4(\text{GREEN} - \text{SWIR1}) - (0.25 \cdot \text{NIR} + 2.75 \cdot \text{SWIR2})$
NDWI	Water	$(\text{GREEN} - \text{NIR}) / (\text{GREEN} + \text{SWIR1})$
LSWI	Water/moisture	$(\text{NIR} - \text{SWIR1}) / (\text{NIR} + \text{SWIR1})$
NDPI	Water/phenology	$(\text{SWIR1} - \text{GREEN}) / (\text{SWIR1} + \text{GREEN})$
WRI	Water	$(\text{GREEN} + \text{RED}) / (\text{NIR} + \text{SWIR1})$
NDVI	Vegetation	$(\text{NIR} - \text{RED}) / (\text{NIR} + \text{RED})$
EVI2	Vegetation	$2.5 \cdot (\text{NIR} - \text{RED}) / (\text{NIR} + 2.4 \cdot \text{RED} + 1)$
SAVI	Vegetation	$1.5 \cdot (\text{NIR} - \text{RED}) / (\text{NIR} + \text{RED} + 0.5)$
GCVI	Vegetation	$(\text{NIR} / \text{GREEN}) - 1$
MSAVI	Vegetation	$0.5 \cdot [2\text{NIR} + 1 - \sqrt{((2\text{NIR} + 1)^2 - 8(\text{NIR} - \text{RED}))}]$
NDMI	Vegetation	$(\text{NIR} - \text{SWIR1}) / (\text{NIR} + \text{SWIR1})$
NDAVI	Vegetation	$(\text{NIR} - \text{BLUE}) / (\text{NIR} + \text{BLUE})$
NDBI	Soil/built	$(\text{SWIR1} - \text{NIR}) / (\text{SWIR1} + \text{NIR})$
BSI	Soil	$[(\text{SWIR1} + \text{RED}) - (\text{NIR} + \text{BLUE})] / [(\text{SWIR1} + \text{RED}) + (\text{NIR} + \text{BLUE})]$
SBI	Soil	$\sqrt{(\text{RED}^2 + \text{NIR}^2)}$
NDTI	Soil	$(\text{SWIR1} - \text{SWIR2}) / (\text{SWIR1} + \text{SWIR2})$
EBBI†	Thermal hybrid	$(\text{SWIR1} - \text{NIR}) / (0.36 \cdot \text{SWIR1} + 0.47 \cdot \text{NIR} + 0.17 \cdot \text{THERMAL})$

† EBBI omitted when thermal band unavailable (years without L2SP processing).

5. Dry-Season Median Composite (v2.0)

5.1 Compositing Strategy

All scenes within 1 November of year Y to 31 May of year Y+1 are collected from the merged five-mission Landsat archive. The pixel-wise median is computed across all valid observations for each band and index independently:

$$C(x,b) = \text{median}_t \{ I_t(x,b) : I_t(x) \text{ valid} \}$$

This produces a spatially homogeneous, sensor-agnostic composite without inter-scene artefacts.

⚠ NoData propagation

Pixels with no valid observation in any scene remain masked (NoData) in the composite. This mask propagates to all GEE output rasters. In the R post-processing pipeline, these pixels are encoded as class=0 (not as NA) due to GEE export behaviour, which requires explicit handling before change detection (see Section 15.3).

5.2 Feature Stack

Group	Max bands	Variables
Landsat median indices	6–20	SR bands (BLUE–SWIR2) ± THERMAL + 13–14 indices
Sentinel-1 (post-2015)	3	S1_VV, S1_VH, S1_VVH_ratio (seasonal median, dB)
TerraClimate (annual mean)	5	tmmn, tmmx, def, vap, soil (bilinearly resampled to 30 m)
Terrain (static)	4	srtm_elevation, srtm_slope, srtm_roughness, srtm_tw

6. Class Schema Definition

6.1 WorldCover Majority Resampling

WorldCover 2020 (10 m) is aggregated to 30 m via mode resampling: `reduceResolution(ee.Reducer.mode()).reproject(30 m)`. Mode resampling is statistically correct for categorical data.

6.2 Fixed JM Reference Year

JM separability statistics are always computed on the fixed JM_YEAR parameter (default: 2020). All years are classified with the same class schema, enabling valid temporal comparison. The JM_YEAR must not be changed between runs.

6.3 Jeffries-Matusita Separability

Per-class statistics are estimated from the JM_YEAR feature stack via a grouped `reduceRegion()` call. Variance is derived as $\text{Var}[X] = E[X^2] - E[X]^2$ from a (2p)-band image containing both original and squared bands — enabling a single server-side call per class. The JM distance between classes i and j is:

$$B_{ij} = (1/8) (\mu_i - \mu_j)^T * \Sigma^{-1} * (\mu_i - \mu_j) + (1/2) * \ln(|\Sigma| / \sqrt{|\Sigma_i| * |\Sigma_j|})$$

$$JM_{ij} = 2 * (1 - \exp(-\max(B_{ij}, 0))) \quad \text{in } [0, 2]$$

6.4 Merge Rules

Classes	Code	Condition	Rationale
WC 50+60 (Built-up+Bare)	55	Always	Spectrally indistinguishable in dry season.
WC 30+40 (Grassland+Cropland)	35	JM<1.8	Post-harvest cropland and dry savanna share near-identical reflectance (JM=0.30 for Ibaji).
WC 10+20 (Tree cover+Shrubland)	15	JM<1.8	Deciduous canopy in dry season may be indistinguishable from shrubland at 30 m resolution.

Final classes for Ibaji: Permanent water bodies (WC80), Herbaceous wetland (WC90), Grassland/Cropland (35), Woody vegetation (15), Non-vegetated/Bare (55).

7. Training Data Preparation

7.1 Self-Supervised Label Strategy

⚠ Critical limitation: circular validation

Training and validation labels both derive from WorldCover 2020. The 70/30 split produces a validation set from the same label source as the training set. Accuracy metrics (OA, PA, UA, Kappa) measure how well the RF reproduces WorldCover 2020 on held-out WorldCover pixels — not how well the classification reflects the true state of the land surface. For publication, independently collected validation points are required.

7.2 Spectral Z-Score Filtering

For each class c and band b , a sample is retained only if $|z_{\{is\}}| \leq \tau$ for ALL bands simultaneously ($\tau=3$ by default) (Richiardi et al., 2025). Implemented server-side via `ee.FeatureCollection.iterate()`.

$$z_{\{isb\}} = (x_{\{isb\}} - \mu_{\{cb\}}) / \sigma_{\{cb\}}$$

7.3 Balanced Sampling and 70/30 Split

Balanced sampling applies first ($\max \text{TARGET_SAMPLES_PER_CLASS}$ per class, $\text{floor} = \max(15, n\text{Trees} \times 0.05)$), then the balanced set is split 70/30. The balanced validation set is required for isotonic calibration in R to function correctly (see Section 13).

8. Classification

8.1 Random Forest

A Random Forest is trained independently for each year on that year's feature stack. No classifier is transferred between years.

Parameter	Value	Rationale
Number of trees	300	Variance reduction vs. computation time
Variables per split	6	$\sim\sqrt{p}$ for $p \approx 20\text{--}29$ predictors
Minimum leaf population	2	Allows deep trees on small samples
Bag fraction	0.70	Standard OOB estimation
Maximum nodes	128	Prevents overfitting on small datasets
Seed	42	Reproducibility (training split seeded to year)

8.2 MULTIPROBABILITY Export

The RF is applied in MULTIPROBABILITY mode, returning a per-class probability for each pixel. GEE's MULTIPROBABILITY returns an array-type image. Exporting array-type images raises 'Cannot export array bands'. Fix: `.arrayFlatten([multiProbBandNames])` converts the array image to a standard N-band float image before export. Band names are `prob_<ClassName>` in ascending class code order.

8.3 Variable Importance Export

After training, `classifier.explain()` is called server-side to extract Gini impurity importance per feature. The result is exported as `variable_importance_YEAR.csv` with columns: feature, importance (raw Gini), importance_norm (normalised by max), rank (1 = most important).

9. Uncertainty and Confidence Estimation

9.1 RF Probability Under-Dispersion

⚠ Critical limitation: raw RF probabilities are uninformative

Raw RF class probabilities are systematically under-dispersed (Niculescu-Mizil & Caruana 2005). With 5 classes, all pixels receive probabilities in the narrow range [0.33–0.39] regardless of classification confidence. The raw uncertainty raster has been removed from v2.0 GEE exports. Isotonic regression calibration in R corrects this (see Section 13).

9.2 Raw Confidence Map (GEE output)

The winning-class probability is exported as a coarse spatial proxy:

$$\text{confidence}(x) = \max_c \{ p_c(x) \} \quad \text{in } [0, 1]$$

Exported as Byte 0–100 (`confidence_YEAR.tif`). Divide by 100 to recover [0,1]. Even a value of 0.9 does not mean 90% probability of correct classification, due to RF under-dispersion.

9.3 Calibrated Confidence and Calibrated Uncertainty (R Module 1)

After isotonic regression calibration (see Section 13.1), two spatially informative products are derived from the calibrated probability distribution $\{p_{\text{cal}_c}(x)\}$ for each pixel x . These two outputs measure fundamentally different aspects of the probability distribution.

9.3.1 Calibrated Confidence

$$\text{calibrated_confidence}(x) = \max_c \{ p_{\text{cal}_c}(x) \} \quad \text{in } [0, 1]$$

Calibrated confidence is the maximum calibrated probability among all classes. It answers: 'How certain is the model about the class it assigned?' A high value means one class clearly dominates the probability distribution. A low value means the model is uncertain about which class to assign.

Exported as Float32 [0,1] in calibrated_confidence_YEAR.tif.

9.3.2 Calibrated Uncertainty

Calibrated uncertainty is the normalised Shannon entropy of the full calibrated probability distribution:

$$H(x) = -\sum_c p_{cal_c}(x) * \log(p_{cal_c}(x)) / \log(n_classes) \quad \text{in} \quad [0, 1]$$

where $n_classes$ is the number of final classes (5 for Ibaji). The normalisation by $\log(n_classes)$ maps the entropy to [0,1] regardless of the number of classes. $H=0$ when all probability mass is on one class (complete certainty). $H=1$ when probability mass is equally distributed across all classes (maximum uncertainty).

Exported as Float32 [0,1] in calibrated_uncertainty_YEAR.tif.

9.3.3 Distinction between confidence and uncertainty

Calibrated confidence and calibrated uncertainty are NOT simply inverse functions of each other. They capture different structural properties of the probability distribution. Consider three probability vectors for a 5-class problem:

Distribution	p_cal	Confidence	Uncertainty H	Interpretation
A: one clear winner	[0.80, 0.10, 0.05, 0.03, 0.02]	0.80	low	Model is certain. One class dominates.
B: two competitors	[0.60, 0.30, 0.05, 0.03, 0.02]	0.60	medium-high	Model uncertain between 2 classes. Entropy penalises the strong runner-up.
C: diffuse	[0.60, 0.10, 0.10, 0.10, 0.10]	0.60	medium-low	Same confidence as B, but uncertainty is LOWER because probability is not competing with a single alternative.
D: uniform	[0.20, 0.20, 0.20, 0.20, 0.20]	0.20	1.0 (max)	Model has no preference.

Case B and C have identical confidence (0.60) but different uncertainty. Case B is ecologically important for Ibaji: pixels at the water/wetland boundary often have high probability for both Permanent water bodies and Herbaceous wetland, creating a strong runner-up. These pixels have lower confidence AND higher uncertainty than spectrally ambiguous pixels spread across multiple classes.

Spatial interpretation: Calibrated confidence identifies where the assigned class is secure. Calibrated uncertainty identifies which confusion dominates spatially — which pair of classes competes at that location. These are complementary spatial diagnostics.

9.3.4 What confidence and uncertainty cannot detect

⚠ **Confident-but-wrong errors are invisible to both metrics**

Both calibrated confidence and calibrated uncertainty identify spectral uncertainty — where the RF is at the edges of its feature space. They cannot detect systematic classification errors where the model is confident but wrong. This occurs when the training label (WorldCover 2020) has a spatial bias in a specific region: the RF learns that bias with high confidence and applies it consistently. A pixel of former agricultural land that was classified as Grassland by WorldCover 2020 but was actually flooded wetland in 1990 will be classified as Grassland with high confidence and low uncertainty in 1990 — and neither metric will flag the error. Only spatially stratified independent validation points can reveal this category of error.

10. Accuracy Assessment (GEE)

Metric	Formula	Note
Overall Accuracy (OA)	correct / total	Balanced val set — does not reflect landscape OA
Kappa	$(OA - p_e)/(1 - p_e)$	Agreement corrected for chance
Balanced Accuracy (BA)	mean(PA_c) across classes	Most robust metric for imbalanced classes
Producer's Accuracy (PA_c)	$TP_c / (TP_c + FN_c)$	PA=0 for classes absent from training
User's Accuracy (UA_c)	$TP_c / (TP_c + FP_c)$	
F1-score_c	$2 \cdot PA_c \cdot UA_c / (PA_c + UA_c)$	Harmonic mean; =0 when PA or UA = 0

11. GEE Outputs (per year)

File	Format	Content
land_cover_YEAR.tif	Int16 GeoTIFF	Classified map; values = final class codes; class=0 encodes NoData (pixels with no valid Landsat observations)
confidence_YEAR.tif	Byte 0–100 GeoTIFF	Raw RF max class probability ×100; divide by 100 for [0,1]
multiprob_YEAR.tif	Byte 0–100 GeoTIFF	N bands (prob_<ClassName>); per-class RF probabilities ×100
accuracy_YEAR.csv	CSV	PA, UA, F1 per class; OA, Kappa, BA overall
validation_points_YEAR.csv	CSV	Per-point: geometry, LC, predicted, correct, prob_* per class
class_lookup_table.csv	CSV	Final class code → class name
jm_distances.csv	CSV	Pairwise JM separability matrix; merge flags
variable_importance_YEAR.csv	CSV	Feature, importance (Gini), importance_norm, rank

12. GEE Implementation Notes

12.1 Asynchronous Architecture

The workflow uses zero synchronous `getInfo()` calls. All startup computations use `evaluate()` callbacks. Total `evaluate()` calls: 9 (1 histogram + 8 per-class JM statistics), all non-blocking.

12.2 NoData Encoding

⚠ **class=0 in GEE exports encodes NoData, not a class**

GEE's `Export.image.toDrive()` encodes masked pixels (NoData) as the raster's default no-data value, which for `Int16` is 0. Since class codes start at 10 (WC10), `class=0` unambiguously identifies pixels with no valid Landsat observations. The R pipeline must mask `class=0` pixels before any change comparison (see Section 15.3, Stream B NoData fix v2.2).

13. R Module 1 — RF Probability Calibration

Module 1 corrects the systematic under-dispersion of raw RF probabilities using isotonic regression, producing spatially informative confidence and uncertainty maps that are used by all downstream modules.

13.1 Isotonic Regression

Isotonic (monotone non-decreasing) regression maps raw RF probabilities to empirically observed correct-classification frequencies on the validation sample (Zadrozny & Elkan 2002).

For each class c , the calibration is a binary classification problem:

$$y_c = 1 \quad \text{if the validation point belongs to class } c, \quad \text{else } y_c = 0$$

The raw RF probability p_c is the predictor. The Pool-Adjacent-Violators (PAV) algorithm fits the optimal non-decreasing step function:

$$p_{cal_c} = f_c(p_{raw_c}) \quad \text{such that} \quad E[f_c(p)] \approx P(\text{true class} = c \mid \text{RF_prob}_c = p)$$

After per-class calibration, probabilities are re-normalised to sum to 1 per pixel:

$$p_{cal_c}(x) := p_{cal_c}(x) / \sum_c p_{cal_c}(x)$$

The calibrated confidence and calibrated uncertainty are then derived as described in Section 9.3.

13.2 Why balanced validation is required

With an unbalanced validation set (e.g. rare classes = 5% of points), the binary calibration sees $y_c=1$ only 5% of the time regardless of the RF probability value. The isotonic function collapses to the global class prevalence, making all calibrated probabilities near zero after normalisation. The balanced validation set (equal class frequency by design) gives each class approximately 20% prevalence in the binary problem, enabling the calibration function to detect the relationship between RF probability and classification correctness.

13.3 Application to multiprob raster

The fitted calibration functions are applied pixel-wise to `multiprob_YEAR.tif` (N-band Float [0,1]). The output is re-normalised and written to disk. Calibrated confidence and calibrated uncertainty are derived and exported as separate Float32 rasters.

13.4 Module 1 Outputs

File	Format	Content
<code>calibrated_confidence_YEAR.tif</code>	Float32 [0,1]	Max calibrated class probability per pixel
<code>calibrated_uncertainty_YEAR.tif</code>	Float32 [0,1]	Normalised Shannon entropy of calibrated probabilities
<code>calibration_functions_YEAR.csv</code>	CSV	Isotonic regression lookup table (<code>p_raw</code> , <code>p_cal</code>) per class
<code>calibration_plot_YEAR.png</code>	PNG	Reliability diagrams + entropy distribution before/after calibration

14. R Module 2 — Annual Quality Index

Module 2 assigns each processed year a composite Quality Index (QI) in [0,1] that summarises the reliability of that year's classification. The QI is used as a regression weight in Module 3 (Stream A) and as a visual flag in all time-series plots.

14.1 QI Formula

$$QI = 0.30 * coverage_norm + 0.30 * confidence_norm + 0.30 * BA_norm + 0.10 * S1_bonus$$

Each component is normalised to [0,1] by dividing by the observed range across the full time series.

Component	Source	Description
<code>coverage_norm</code>	<code>land_cover_YEAR.tif</code>	Fraction of pixels with valid classification (non-NA). Intended to penalise years with low Landsat coverage.
<code>confidence_norm</code>	<code>confidence_YEAR.tif</code>	Spatial mean of the raw confidence map.
<code>BA_norm</code>	<code>accuracy_YEAR.csv</code>	Balanced Accuracy from the GEE validation output.
<code>S1_bonus</code>	<code>config.R</code> <code>S1_START_YEAR</code>	Binary: 1 if year $\geq$ <code>S1_START_YEAR</code> (default 2015), else 0.

⚠ Known limitation: coverage component is non-functional

GEE encodes NoData as `class=0` (see Section 12.2). When R reads the raster, `class=0` pixels are valid integers, not NA. Therefore `terra::freq()` counts them as belonging to a class, and the fraction of non-NA pixels is always 1.0 for all years. The coverage component adds nothing to the QI ranking. The correct implementation would use the number of valid Landsat scenes per season, exportable from GEE metadata. This is flagged for a future version.

14.2 Low-Quality Flags

Years with $QI < QI_LOW_THRESHOLD$ (default 0.5) are flagged as `low_quality`. In Module 3 time-series plots, these years are highlighted with grey background bands. In OLS regressions, low-quality years receive lower weight.

14.3 Module 2 Outputs

File	Content
quality_index.csv	Year, all raw components, normalised components, QI, low_quality flag
quality_index.png	Time series of QI with stacked component contributions and low-quality threshold line

15. R Module 3 — Land Cover Change Analysis

Module 3 is the main analytical output module. It operates on all processed years simultaneously and produces two independent change analysis streams: Stream A (confidence-weighted temporal smoothing on complete maps) and Stream B (pixel-level uncertainty-gated change detection on raw maps).

15.1 Pixel Area Computation (v2.3 fix)

⚠ Geographic CRS: pixel area must be computed from latitude-corrected metre conversion

GEE exports rasters in EPSG:4326 (WGS84 geographic). `terra::res()` on a geographic raster returns degrees, not metres. The naive calculation `prod(terra::res(ref))/1e6` gives $\sim 7 \times 10^{-14} \text{ km}^2$ instead of $\sim 0.0009 \text{ km}^2$ (factor $\sim 10^{10}$ error). Fix introduced in v2.3: detect CRS type; if geographic, compute pixel area using latitude-corrected degree-to-metre conversion. This fix affects all absolute area values in km^2 — relative comparisons (%) are unaffected.

```
pix_km2 = (res_x * 111320 * cos(lat_mid * pi/180)) * (res_y * 110540)
          / 1e6
```

15.2 Stream A — Confidence-weighted Temporal Smoothing

Raw classified maps have variable NoData coverage (18.8% AOI in 1985, $\sim 39\%$ for most years, $\sim 0\%$ from 2013). Direct area comparison between years with different NoData coverage would be methodologically incorrect.

The confidence-weighted temporal smoothing produces complete maps (zero NoData) by assigning each pixel the class with the highest accumulated confidence weight over a temporal window:

```
smoothed_class(x,t) = argmax_c sum_{s in [t-h,t,t+h]}
                           (confidence(x,s)/100) * 1[class(x,s) = c]
```

where h is the half-window size (default $h=1$). Calibrated confidence is used where available (Module 1 output); raw confidence as fallback. The resulting smoothed maps have 100% coverage and are pixel-wise comparable across all years.

On these smoothed maps, Stream A computes:

- Area by class and year (km^2 and % of AOI)

- QI-weighted OLS linear trend per class: $\text{area_km}^2(t) = \alpha + \beta \cdot t + \epsilon$, weights = QI(t)
- 95% confidence intervals for each slope estimate
- Transition matrix (first → last year, in km²)
- Stable mask (pixels unchanged across the full series)
- Percentage change relative to first year at three snapshots (initial, mid, final)
- Alluvial (Sankey) diagram showing class flows across three time snapshots

15.3 Stream B — Uncertainty-gated Change Detection

Stream B works on the raw classified maps (not smoothed) to assess what fraction of year-to-year pixel changes is supported by sufficient classification confidence in both years.

⚠ v2.2 fix: NoData (class=0) excluded before comparison

Before this fix, Stream B compared raw classified maps directly. class=0 (NoData pixels with no Landsat observations) was treated as a valid class value and generated spurious transitions: 0→class and class'0. For the pair 1985→1986, up to 86% of all apparent 'changes' were false positives caused by the shifting NoData mask. Fix: pixels with class=0 in either year are excluded from comparison. Only the intersection of valid pixels (class>0 in BOTH years) is analysed. The output includes pct_valid_overlap to document how much of the AOI is genuinely comparable.

For each pair of consecutive years (t, t+1), each valid pixel is assigned to one of three categories:

```
stable(x)          = 1  if class(x,t) = class(x,t+1)  [within valid
                                                         mask]

certain_change(x) = 1  if class(x,t) != class(x,t+1)  AND  conf(x,t)
                    >= threshold AND  conf(x,t+1) >= threshold

uncertain_change(x) = 1  if class(x,t) != class(x,t+1)  AND
                    (conf(x,t) < threshold OR  conf(x,t+1) < threshold)
```

The key output metric is:

```
fraction_certain = n_certain_change / (n_certain_change +
                                       n_uncertain_change)
```

i Stream B requires calibrated confidence

With raw RF confidence, fraction_certain is <1% in all years regardless of threshold. This is not a bug — it is the direct consequence of RF under-dispersion: pixels that change class between years are precisely those at the spectral boundaries, which always have the lowest confidence values. With calibrated confidence (Module 1) and a threshold of 50%, fraction_certain rises to a mean of 35.8% across all year pairs — a meaningful and interpretable signal.

Parameter	Default	Description
CONF_THRESHOLD	50	Minimum confidence (0–100) required in BOTH years for a change to be classified as 'certain'. Meaningful only with calibrated confidence.

USE_CALIBRATED_CONF	TRUE	If TRUE and calibrated_confidence_YEAR.tif exists, it is used instead of raw confidence.
SMOOTH_HALF_WINDOW	1	Half-window size for Stream A temporal smoothing (± 1 year by default).

15.4 Module 3 Outputs

File	Content
smoothed_YEAR.tif (per year)	Int16 GeoTIFF — confidence-weighted smoothed classification (100% coverage)
area_by_class_year.csv	Area (km ² and %) per class per year, with QI
trend_by_class.csv	OLS slope, CI, R ² , p-value, % change per class
transition_{Y1}_{Y2}.csv	Transition matrix (km ²) from first to last year (smoothed maps)
change_first_last.tif	Encoded transition raster: value = from_code*10000 + to_code
stable_mask.tif	Binary: 1 = pixel stable throughout the full series
uncertainty_reliability.csv	Per year-pair: n_nodata_excluded, n_valid_overlap, pct_valid_overlap, fraction_certain, pct_valid_changed
area_timeseries.png	Time series of class areas
stacked_area.png	Stacked proportional composition by year
trends.png	OLS slope $\pm$ CI per class (forest plot)
transition_heatmap.png	Transition matrix heatmap
uncertainty_reliability.png	Stacked bar chart: stable/certain/uncertain/nodata per year-pair
alluvial_transitions.png	Alluvial diagram: class flows across three snapshots
pct_change_by_class.png	Percentage change relative to first year at mid and final snapshot
pct_change_by_class.csv	Underlying data for the percentage change plot

16. R Module 4 — Variable Importance Over Time

Module 4 aggregates the variable importance exported by GEE across all processed years and provides comparative visualisations that reveal which features drive the RF and how the feature ranking changes with the introduction of Sentinel-1.

16.1 Aggregation

The variable_importance_YEAR.csv files from GEE are loaded for all available years. Features are assigned to thematic groups defined in config.R (Water, Vegetation, Soil_Bare, Climate, Terrain). For each feature, Module 4 computes: mean rank across years, median rank, number of years present (e.g. S1_VV is present only for years ≥ 2015), and importance normalised by the series maximum.

16.2 Key Findings for Ibaji

Feature	Mean rank	Type	Availability	Note
S1_VV	1.8	SAR backscatter	≥2015 only	Dominant when available; distinguishes double-bounce (flooded veg.) from specular (open water)
srtm_twi	3.1	Terrain	All 32 years	Top-3 in every year. Topographic position proxies inundation probability.
tc_tmmx_mean	4.2	Climate	All years	Max temperature determines evapotranspiration and vegetation stress.
srtm_elevation	5.0	Terrain	All years	
S1_VH	5.3	SAR backscatter	≥2015 only	
NDVI	12–30	Optical	All years	Varies widely; lower than expected for wetland mapping.
MNDWI	9–33	Optical	All years	Low rank despite being the primary water index. TWI predicts water presence more stably.
srtm_slope	~36	Terrain	All years	Consistently last or penultimate. Flat floodplain: slope provides no discriminative information.

16.3 Module 4 Outputs

File	Content
vip_table.csv	Full aggregated importance table (feature × year)
vip_top_features.csv	Summary: mean rank, median rank, n_years per feature
vip_heatmap.png	Feature × year heatmap of normalised importance (coloured by group)
vip_top10_timeseries.png	Time series of mean rank for top-10 features
vip_rank_stability.png	Spearman ρ of rank vectors between consecutive years (stability indicator)
vip_pre_post_s1.png	Barplot: change in mean rank for each feature pre- vs post-S1 introduction

17. Known Limitations

Limitation	Severity	Mitigation
Circular validation	Critical	Training and validation labels both derive from WorldCover 2020. Accuracy metrics measure RF reproducibility of WorldCover, not true land cover accuracy. Required: independently collected validation points for ≥ 3 epochs.
Landsat scene scarcity 1985–2003	High	Coverage ranges from 18.8% (1985) to ~60% (2012). Smoothing fills gaps but propagates information from adjacent years, not independent observations.
RF probability under-dispersion	High	Corrected by isotonic calibration (Module 1) for years where multiprob rasters are available and the validation set is sufficient.
Sentinel-1 absent before 2014	High	The primary discriminator for water/flooded vegetation is unavailable for the first 30 years of the series. Optical-only classification has higher water/wetland confusion in this period.
class=0 NoData encoding	Medium	Stream B (v2.1 and earlier) treated NoData as a class, generating up to 86% false positives. Fixed in v2.2. Stream A (smoothed maps) is unaffected.
pix_km2 geographic CRS error	Medium	Area values in km ² were wrong by a factor of $\sim 10^{10}$ when computed from geographic-degree rasters. Fixed in v2.3. Relative (%) comparisons were unaffected.
QI coverage component non-functional	Low	GEE encodes NoData as class=0, so coverage is always 1.0. Flagged for future fix.
WorldCover label temporal distance	Medium	WorldCover 2020 labels applied to 1985–2024. Areas with genuine land-use change between 1985 and 2020 may be systematically misclassified in historical years.
srtm_slope useless in flat floodplain	Low	Slope is consistently last in VIP rankings. Removing it would reduce noise. Retained for cross-site reproducibility.

18. AI Assistance Documentation

In compliance with emerging standards from COPE (Committee on Publication Ethics), IEEE, Nature Portfolio, and MDPI for the disclosure of AI tools in scientific research, this section documents the use of AI assistance in the development of the algorithm and analysis described in this ATBD.

18.1 Tool Used

Tool	Provider	Access method	Period of use
Claude (claude-sonnet-4-5, claude-sonnet-4-6)	Anthropic PBC	claude.ai (web interface, Project mode)	2025

18.2 Scope and Nature of AI Assistance

The AI assistance in this project extended beyond language editing or manuscript polishing. It was used as an interactive co-development tool across the following categories:

Category	Examples	Disclosure level
Algorithm design	GEE pipeline architecture, JM separability implementation, isotonic calibration strategy, Stream A/B change detection design	Methods section
Code generation	GEE JavaScript script (full pipeline), R modules 1–4 (full implementation)	Methods section
Debugging	NoData bug in Stream B (v2.2), unit error in pix_km2 (v2.3), alluvial plot column naming error, Scale_fill_manual setNames fix	Methods section
Methodological analysis	Identification of RF under-dispersion and its implications, diagnosis of QI coverage component non-functionality, VIP ranking analysis, fraction_certain interpretation	Methods section
Interpretation of results	Analysis of transition matrix, identification of water→bare transition as ecologically significant, distinction between spectral uncertainty and systematic bias	Methods section
Document preparation	ATBD drafting and revision, presentation structure, briefing materials	Acknowledgments

18.3 Policy Compliance

AI tools cannot be listed as authors.

Authorship requires legal and intellectual responsibility that AI systems cannot assume. All outputs generated with AI assistance were reviewed, validated, and interpreted by the human authors, who take full responsibility for the scientific content of this document and the associated code.

The human authors verified all AI-generated content.

This includes: reviewing all code for correctness and appropriateness; re-running analyses and confirming outputs; identifying and correcting errors in AI-generated code (the three bugs fixed in v2.2 and v2.3 were initially present in AI-generated implementations); and exercising independent scientific judgment in the interpretation of results.

Accuracy of AI-generated content is the authors' responsibility.

The use of AI assistance does not alter the authors' responsibility for the accuracy of the methodology, the correctness of the code, or the validity of the scientific conclusions.

19. References

Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32.

Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1), 37–46.

ESA (2021). WorldCover 10 m 2020 v100. European Space Agency. <https://doi.org/10.5281/zenodo.5571936>

- Hermosilla, T., Wulder, M. A., White, J. C., Coops, N. C., & Hobart, G. W. (2018). Disturbance-informed annual land cover classification maps of Canada's forested ecosystems for a 29-year Landsat time series. *Canadian Journal of Remote Sensing*, 44(1), 67–84.
- Jeffries, H., & Matusita, K. (1951). Fundamental aspects of the problem of classification. *Annals of the Institute of Statistical Mathematics*, 10, 211–221.
- Niculescu-Mizil, A., & Caruana, R. (2005). Predicting good probabilities with supervised learning. *Proceedings of ICML*, 625–632.
- Richiardi, C., Siniscalco, C., Garbarino, M. et al. (2025). Unravelling decades of habitat dynamics in protected areas: A hierarchical approach applied to the Gran Paradiso National Park (NW Italy). *Environ Monit Assess* 197, 1216. <https://doi.org/10.1007/s10661-025-14669-0>.
- Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3), 379–423.
- Telegraaf, J., Leenders, B., & Luijendijk, E. (2023). Global mapping of annual wetland dynamics using Landsat. *Hydrology and Earth System Sciences*, 27, 3145–3159.
- Yamazaki, D. et al. (2019). MERIT Hydro: A high-resolution global hydrography map. *Water Resources Research*, 55(6), 5053–5073.
- Zadrozny, B., & Elkan, C. (2002). Transforming classifier scores into accurate multiclass probability estimates. *Proceedings of KDD*, 694–699.